

- 1 The sum of the first 3 terms of a geometric progression is 8. The sum of the first 6 terms of the same progression is 72.
What is the common ratio of the progression?

A 2
B 3
C 4
D 8
E 9

- 2 The diagram shows a triangle with side lengths $x + 3$, $2x - 1$, and $11 - x$.
Find the complete set of real values of x for which a non-degenerate triangle with these side lengths exists.

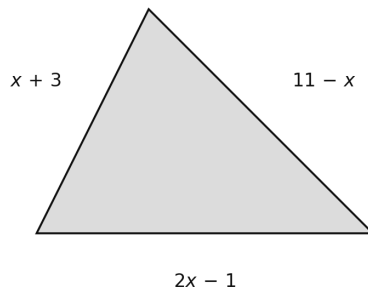


Diagram not to scale

- A $x > \frac{9}{4}$
B $x < \frac{15}{2}$
C $\frac{1}{2} < x < 11$
D $\frac{9}{4} < x < \frac{15}{2}$
E $\frac{9}{4} < x < 11$
F $\frac{1}{2} < x < \frac{15}{2}$
G $\frac{1}{2} < x < \frac{9}{4}$
H $\frac{15}{2} < x < 11$

- 3 The diagram shows a regular hexagon $ABCDEF$ with sides of length a .

A circular arc with centre A is drawn from vertex C to vertex E .

What is the area of the shaded region bounded by the line segments CD and DE and the arc CE ?

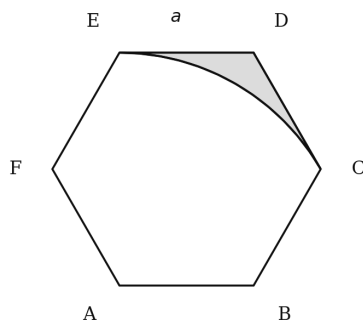


Diagram not to scale

- A** $\left(\frac{\sqrt{3} - \pi}{2} \right) a^2$
- B** $\left(\frac{2\sqrt{3} - \pi}{4} \right) a^2$
- C** $\left(\frac{2\sqrt{3} - \pi}{2} \right) a^2$
- D** $\left(\frac{3\sqrt{3} - \pi}{2} \right) a^2$
- E** $\left(\frac{4\sqrt{3} - \pi}{4} \right) a^2$
- F** $\left(\frac{3\sqrt{3} - 2\pi}{4} \right) a^2$

- 4 The table gives values of a function f .

x	0	1	2	3
$f(x)$	2	3	6	11

It is also known that $f''(x) > 0$ for $0 < x < 3$. The trapezium rule with these four ordinates is used to estimate

$$\int_0^3 f(x) dx.$$

Which statement is correct?

- A 15, underestimate
 - B 15, overestimate
 - C $\frac{31}{2}$, underestimate
 - D $\frac{31}{2}$, overestimate
 - E 16, underestimate
 - F 16, overestimate
- 5 A parabola has the equation $y = x(k - x)$ for some constant $k > 0$.
A triangle is formed by the two points where the parabola meets the x -axis and by the vertex of the parabola.

What is the total area of the regions that lie between the parabolic arc and the sides of the triangle?

- A $\frac{k^3}{24}$
- B $\frac{k^3}{12}$
- C $\frac{k^3}{8}$
- D $\frac{k^3}{6}$
- E $\frac{k^3}{4}$

- 6 Find the area of the finite region defined by the inequalities:

$$y \geq |2x + 4|$$

$$y \leq 8$$

- A 8
- B 16
- C 24
- D 32
- E 48

- 7 Find the sum of the squares of the real roots of the equation

$$x\sqrt{1} = \frac{1}{3}$$

- A $\frac{1}{9}$
- B $\frac{1}{3}$
- C $\frac{7}{9}$
- D 1
- E 2

- 8 A company's annual profit is modelled to decrease by a fixed percentage, $p\%$, at the end of each year.

The total profit predicted for the first two years is 36% of the total profit predicted if the company were to operate forever under these conditions.

What is the value of p ?

- A 20
- B 36
- C 40
- D 60
- E 64
- F 80

- 9 A bag contains 7 gold tokens and 5 silver tokens. The tokens are identical apart from their colour. A player draws tokens one at a time at random from the bag without replacement.

The player begins with a score of 0.

- Each gold token drawn increases the score by 1 point.
- When a silver token is drawn, the score decreases by 2 points and the game ends immediately.
- If all 7 gold tokens are drawn without drawing a silver token, the game also ends.

What is the probability that the player finishes the game with a score strictly greater than 0?

- A $\frac{35}{396}$
B $\frac{7}{99}$
C $\frac{7}{44}$
D $\frac{5}{22}$
E $\frac{7}{22}$
F $\frac{5}{11}$

- 10 A particle undergoes three successive displacements, $\mathbf{d}_1$, $\mathbf{d}_2$, and $\mathbf{d}_3$, and returns to its starting position. $\mathbf{d}_1$ has magnitude $\sqrt{3}$ units and a bearing of 090° . $\mathbf{d}_2$ has magnitude 1 unit and a bearing of 180° .

What is the bearing of $\mathbf{d}_3$?

- A 060°
B 120°
C 240°
D 300°
E 330°

- 11 What is the coefficient of x^{24} in the expansion of the expression below?

$$((1 - 2x^2)(1 + 2x^2 + 4x^4 + 8x^6))^5$$

- A -245760
B -40960
C -80
D 0
E 20480
F 40960

- 12** The definite integral of a function $f(x)$ over the interval $0 \leq x \leq 6$ is given by $\int_0^6 f(x) \, dx = -12$.
The area of the region bounded by the curve $y = f(x)$, the x -axis, and the lines $x = 0$ and $x = 6$ that lies above the x -axis is 5.
What is the area of the region bounded by the curve, the x -axis, and the lines $x = 0$ and $x = 6$ that lies below the x -axis?

- A** 5
- B** 7
- C** 12
- D** 17
- E** 22

- 13** What is the value of the following sum?

$$\sum_{n=1}^{63} \log_4 \left(\frac{n^2 + n}{n^2 + 2n + 1} \right)$$

- A** -6
- B** -3
- C** 0
- D** 3
- E** 6

- 14** A straight line with a negative gradient passes through the point $(4, 3)$.
The line forms a triangle with the positive x -axis and positive y -axis.
The line $y = 2x$ divides this triangle into two smaller triangles.
The area of the triangle with a side on the x -axis is three times the area of the triangle with a side on the y -axis.
What is the area of the triangle formed by the straight line and the positive coordinate axes?

- A** $\frac{289}{12}$
- B** 27
- C** $\frac{121}{4}$
- D** $\frac{75}{2}$
- E** $\frac{289}{6}$

- 15 How many distinct real solutions does the following equation have?

$$\ln(x - 5) + \ln(2 - x) = \ln(2)$$

- A 0
- B 1
- C 2
- D 3
- E 4
- F 7

- 16 A sector of a circle has a fixed perimeter P .
Which expression gives the radius r that maximises the area of the sector?

- A $\frac{P}{2+\pi}$
- B $\frac{P}{4}$
- C $\frac{P}{\pi}$
- D $\frac{P}{3}$
- E $\frac{P}{2}$

- 17 A sequence is defined by $u_1 = 2$ and the recurrence relation

$$u_{n+1} = \ln(e^{u_n-1} - k)$$

for $n \geq 1$, where k is a real constant.

The sequence terminates if $e^{u_n-1} - k \leq 0$.

Find the set of values of k for which the sequence has exactly two strictly positive terms.

- A $0 \leq k < e - 1$
- B $0 \leq k < \frac{e}{e+1}$
- C $0 < k \leq e - 1$
- D $0 \leq k \leq e - 1$
- E $\frac{e^2 - 1}{2} \leq k < e^2 - 1$

- 18** Two rectangles, A and B, have dimensions that depend on a variable x , which changes with time.

The side lengths of rectangle A are $(\log_2(8) + 3x)$ and $(12 - x)$.

The side lengths of rectangle B are $(\log_3(27) + x)$ and $(6 - x)$.

At a certain instant, the perimeter of rectangle A is 42. At this instant, x is decreasing.

What is the value of the ratio $\frac{\text{Area of A}}{\text{Area of B}}$ at this instant, and is this ratio increasing or decreasing?

- A** The ratio is 3 and it is decreasing.
- B** The ratio is 3 and it is increasing.
- C** The ratio is 6 and it is decreasing.
- D** The ratio is 6 and it is increasing.
- E** The ratio is $\frac{1}{6}$ and it is decreasing.
- F** The ratio is $\frac{1}{6}$ and it is increasing.

- 19** Find the x -coordinate of the stationary point of the curve with equation:

$$y = \frac{x^2 - 4x}{x^2 - 16}(x + 4)^2$$

- A** -4
- B** -2
- C** 0
- D** 2
- E** 4

- 20** A particle is dropped from a height of 10 m. After each bounce, it reaches a maximum height that is a constant fraction, r , of the maximum height of the previous bounce, where $0 < r < 1$.

The particle comes to rest after travelling a total vertical distance of 50 m.

What is the value of r ?

- A** $r = \frac{1}{2}$
- B** $r = \frac{3}{5}$
- C** $r = \frac{2}{3}$
- D** $r = \frac{5}{7}$
- E** $r = \frac{4}{5}$

21 Which of the following statements is/are correct?

1. $2^{\log_3 5} = 5^{\log_3 2}$

2. $\frac{\sqrt{75} - \sqrt{12}}{\sqrt{27}} = 1$

3. $\frac{\log_{10} 8}{\log_{10} 2} = \log_{10} 4$

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

22 Find the sum of all solutions to the equation

$$\ln(-\sin x) + \ln(-\cos x) = -\ln 4$$

in the interval $0 \leq x \leq 2\pi$.

A 0

B $\frac{\pi}{2}$

C π

D $\frac{5\pi}{2}$

E 3π

- 23** The index, $I(f)$, of a polynomial $f(x)$ is defined by the formula:

$$I(f) = d + s + |f(-1)|$$

where d is the degree of $f(x)$, and s is the number of distinct real stationary points of $f(x)$.

Consider the two polynomials:

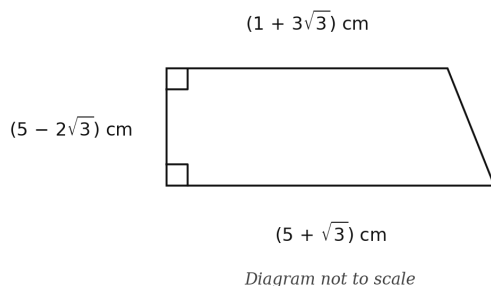
$$f_1(x) = x^3 - 3x + 1$$

$$f_2(x) = x^3 - 5$$

What is the value of $I(f_1) + I(f_2)$?

- A** 6
 - B** 8
 - C** 16
 - D** 17
 - E** 18
 - F** 20
- 24** A rock sample contains a radioactive isotope X and its stable decay product Y. The ratio of the number of nuclei of X to the number of nuclei of Y is 1:3.
The half-life of X is T.
Assuming that there was no Y in the sample initially and that all nuclei of Y were produced by the decay of X, which of the following is the age of the sample?
- A** 1.5 T
 - B** 2 T
 - C** 3 T
 - D** 4 T
 - E** T

- 25** The diagram shows a right-angled trapezium with parallel horizontal sides of length $(1 + 3\sqrt{3})$ cm and $(5 + \sqrt{3})$ cm, and a perpendicular height of $(5 - 2\sqrt{3})$ cm.
- What is the area of the trapezium?



- A** $(3 + 4\sqrt{3}) \text{ cm}^2$
- B** $(6 + 8\sqrt{3}) \text{ cm}^2$
- C** $(27 + 4\sqrt{3}) \text{ cm}^2$
- D** $(21 - 4\sqrt{3}) \text{ cm}^2$
- E** $(3 + 16\sqrt{3}) \text{ cm}^2$
- F** $(9 + 2\sqrt{3}) \text{ cm}^2$
- 26** A sequence is defined by the general term

$$u_n = \frac{9^{n+2}}{3^{2n+1}}$$

for integers $n \geq 1$.

What is the value of the sum of the first 10 terms of the sequence?

- A** $\frac{10}{27}$
- B** 27
- C** 30
- D** 270
- E** 810
- 27** Find the area of the finite region bounded by the curve with equation $4y + |3x| = 24$ and the x -axis.
- A** 24
- B** 48
- C** 96
- D** 144
- E** 192
- F** 288